

Hospital Management System SQL Project

Patient & appointment analytics for a MySQL-backed hospital operations database — from raw records to operational insight.

MySQL • MySQL Workbench

Healthcare Domain

Database: pgj_hospital

19 Verified Tasks

2

CORE TABLES

19

SQL TASKS

10

BASIC LEVEL

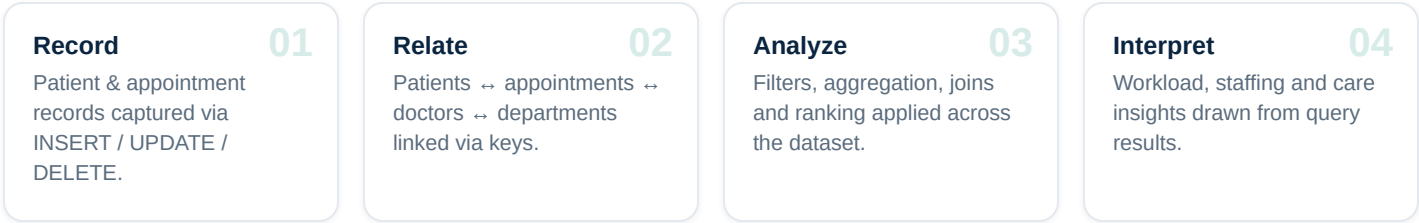
9

INTERMEDIATE-
ADVANCED

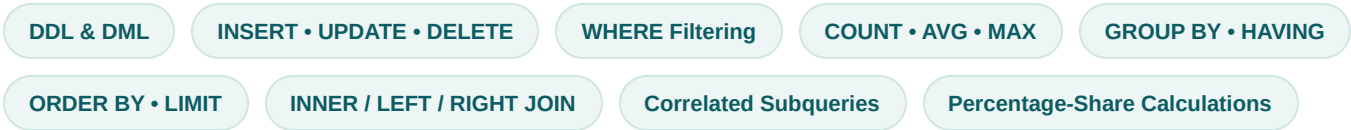
● Project Aim

Use SQL to manage and analyze hospital operational data across patients, appointments, doctors and departments — turning raw records into staffing, workload and care insights.

● Workflow



● SQL Toolkit



● Skill Progression

Stage	Techniques Covered
FOUNDATION	INSERT • UPDATE • DELETE • WHERE filtering • basic joins
INTERMEDIATE	GROUP BY • HAVING • aggregation • multi-column joins
ADVANCED	Correlated subqueries • ranking (ORDER BY + LIMIT) • percentage-share calculations

● Relational Model

PATIENTS	TABLE	APPOINTMENTS	TABLE
Patient_ID	PRIMARY KEY	Appointment_ID	PRIMARY KEY
Patient_Name		Patient_ID	FOREIGN KEY
Age		Doctor_Name	
Gender		Appointment_Date	
Admission_Date		Department	

Patients

1 → N

Appointments

N → 1

Doctor & Department

● Business Analysis Roadmap — 19 Tasks

10

SECTION A • BASIC LEVEL

- Record management (INSERT / UPDATE / DELETE)
- Filtering & NULL handling
- Basic joins
- Counting & distinct values
- Simple updates & deletes

9

SECTION B • INTERMEDIATE → ADVANCED

- Doctor workload analysis
- Multi-department analysis
- Latest visits & missing appointments
- Aggregation across joins
- Ranking & percentage share

Level 1 · Basic SQL Tasks

Tasks 1–3 of 10

01

Data Manipulation

✓ Verified

Insert a new patient record into the **Patients** table with the following details: Patient_ID = 'P101', Patient_Name = 'Chris Johnson', Age = 54, Gender = 'Male', Admission_Date = '2024-08-20'.

QUERY

```
INSERT INTO patients VALUES
(
  'P101', 'Chris Johnson', 54, 'Male', '2024-08-20'
);
```

RESULT

9 13:42:39 insert into patients values ('P101','Chris Johnson',54,'Male','2024-08-20') 1 row(s) affected

02

Data Manipulation

✓ Verified

Update the appointment date for Appointment_ID = 'A003' as it has been rescheduled to '2024-09-15'.

QUERY

```
UPDATE appointments
SET
  appointment_date = '2024-09-15'
WHERE
  appointment_id = 'A003';
```

RESULT

10 13:43:24 UPDATE appointments SET appointment_date = '2024-09-15' WHERE appointment_id = 'A003'

03

Joins

✓ Verified

List all patient names along with their corresponding doctor and appointment date.

QUERY

```
SELECT
  p.Patient_name, a.Doctor_name,
  a.Appointment_date
FROM
  patients AS p
  INNER JOIN appointments AS a
    ON p.patient_id = a.patient_id;
```

RESULT

Result Grid			
Filter Rows:		Export:	Wrap Cell Content:
Patient_name	Doctor_name	Appointment_date	
▶ Melissa Jones	Lisa Moreno	14-09-2023	
Austin Martin	Dr. Matthew Mccarty PhD	07-09-2023	
Ryan George	Travis Jones	23-10-2023	
Steven Smith	Eric Yoder	20-12-2023	
Steven Smith	Ronald Harvey	09-03-2024	
Lisa Lawson	Craig White	21-10-2023	
Brian Brown	Daniel Bailey	22-08-2024	
Brian Brown	Marcus Levine	06-05-2024	
Brian Brown	Emily Blankenship	28-02-2024	

04

Joins & NULL Handling

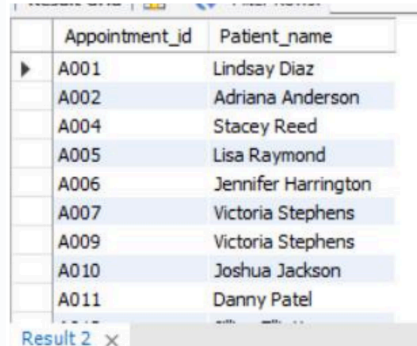
✓ Verified

List all appointments with corresponding patient names, displaying NULL for patients not found in the Patients table.

QUERY

```
SELECT
  a.Appointment_id, p.Patient_name
FROM
  patients AS p
  RIGHT JOIN appointments AS a
    ON p.patient_id = a.patient_id;
```

RESULT



Appointment_id	Patient_name
A001	Lindsay Diaz
A002	Adriana Anderson
A004	Stacey Reed
A005	Lisa Raymond
A006	Jennifer Harrington
A007	Victoria Stephens
A009	Victoria Stephens
A010	Joshua Jackson
A011	Danny Patel

05

Aggregation

✓ Verified

Retrieve the total number of male and female patients in the hospital.

QUERY

```
SELECT
  Gender, COUNT(patient_id) AS Number
FROM
  patients
GROUP BY
  gender;
```

RESULT



Gender	Number
Male	53
Female	49

06

Distinct Values

✓ Verified

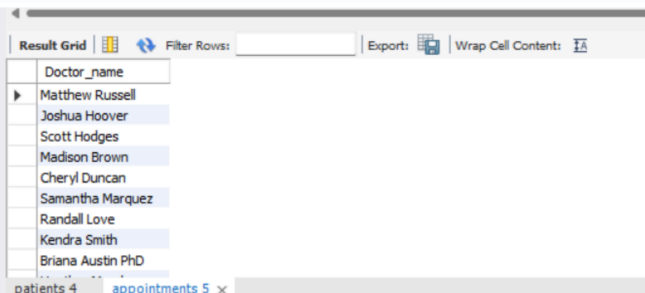
Display all distinct doctor names and all distinct patient names from the respective tables.

QUERY

```
SELECT DISTINCT Patient_name FROM patients;

SELECT DISTINCT Doctor_name FROM appointments;
```

RESULT



Doctor_name	Patient_name
Matthew Russell	Lindsay Diaz
Joshua Hoover	Adriana Anderson
Scott Hodges	Stacey Reed
Madison Brown	Lisa Raymond
Cheryl Duncan	Jennifer Harrington
Samantha Marquez	Victoria Stephens
Randall Love	Victoria Stephens
Kendra Smith	Joshua Jackson
Briana Austin PhD	Danny Patel

07

Subqueries

✓ Verified

Retrieve the details (name and age) of the oldest patient in the hospital.

QUERY

```
SELECT
  Patient_name, Age
FROM
  patients
WHERE
  age = (SELECT MAX(age) FROM patients);
```

RESULT

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
Patient_name	Age			
Anna Carter	100			
Donald Wright	100			

08

Data Manipulation

✓ Verified

Update the department of the doctor named 'Matthew Russell' to 'Radiology'.

QUERY

```
UPDATE appointments
SET
  department = 'Radiology'
WHERE
  doctor_name = 'Matthew Russell';
```

RESULT

✓	17 13:47:05	UPDATE appointments SET	department = 'Radiology' WHERE	doctor_name = 'Matthew Russell'
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09

Data Manipulation

✓ Verified

Delete all appointments belonging to the 'Neurology' department.

QUERY

```
DELETE FROM appointments
WHERE
  department = 'Neurology';
```

RESULT

✓	18 13:47:47	DELETE FROM appointments WHERE	department = 'Neurology'
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10

Subqueries

✓ Verified

Update the department to 'Radiology' for all appointments where the patient's age is greater than 60.

QUERY

```
UPDATE appointments
SET
  department = 'Radiology'
WHERE
  patient_id IN (
    SELECT patient_id
    FROM patients
    WHERE age > 60
  );
```

RESULT

✓ 19 13:48:10 UPDATE appointments SET department = 'Radiology' WHERE patient_id IN (SELECT patient_id FR...

Level 2 · Intermediate → Advanced

Tasks 1–2 of 9

01

Ranking

✓ Verified

Retrieve the doctor who has the most appointments scheduled.

QUERY

```
SELECT
    Doctor_name
FROM
    appointments
GROUP BY
    doctor_name
ORDER BY
    COUNT(appointment_id) DESC
LIMIT 1;
```

RESULT

Result Grid	Filter Rows:
Doctor_name	
▶ Matthew Russell	

02

Multi-Department Analysis

✓ Verified

List the names of patients who have appointments in more than one department.

QUERY

```
SELECT
    p.patient_id, p.patient_name,
    COUNT(DISTINCT a.department) AS
    Appointments
FROM
    patients AS p
    INNER JOIN appointments AS a
        ON p.patient_id = a.patient_id
GROUP BY
    p.patient_id, p.patient_name
HAVING
    COUNT(DISTINCT a.department) > 1;
```

RESULT

Result Grid	Filter Rows:
Patient_name	Appointments
▶ Abigail Smith	2
Adriana Anderson	2
Arthur Lloyd	2
Brian Brown	2
Darryl Scott	2

03

Latest Visits

✓ Verified

Retrieve the most recent appointment date for each doctor.

QUERY

```
SELECT
    Doctor_name, MAX(appointment_date) AS
    Recent_date
FROM
    appointments
GROUP BY
    doctor_name;
```

RESULT



	Doctor_name	Recent_date
▶	Matthew Russell	15-01-2024
	Joshua Hoover	09-04-2024
	Scott Hodges	30-11-2023
	Madison Brown	18-05-2024
	Cheryl Duncan	18-03-2024

04

Missing Appointments

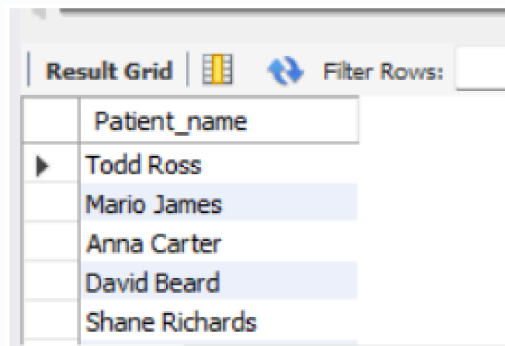
✓ Verified

List patients who do not have any scheduled appointments.

QUERY

```
SELECT
    p.Patient_name
FROM
    patients AS p
    LEFT JOIN appointments AS a
        ON p.patient_id = a.patient_id
GROUP BY
    p.patient_name
HAVING
    COUNT(a.appointment_date) = 0;
```

RESULT



	Patient_name
▶	Todd Ross
	Mario James
	Anna Carter
	David Beard
	Shane Richards

05

Aggregation

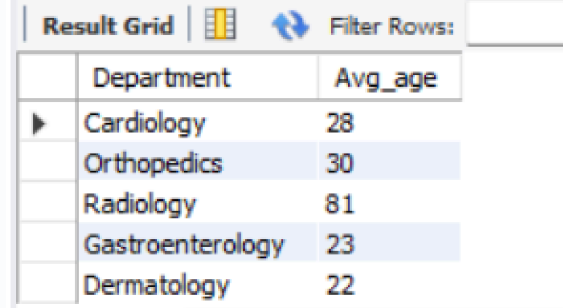
✓ Verified

Calculate the average age of patients for each department.

QUERY

```
SELECT
  a.Department, ROUND(AVG(p.age), 0) AS Avg_age
FROM
  patients AS p
  INNER JOIN appointments AS a
    ON p.patient_id = a.patient_id
GROUP BY
  a.department;
```

RESULT



	Department	Avg_age
▶	Cardiology	28
	Orthopedics	30
	Radiology	81
	Gastroenterology	23
	Dermatology	22

06

Doctor Workload

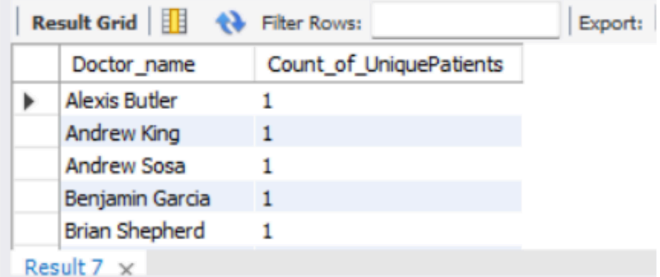
✓ Verified

List all doctors along with the number of unique patients they have attended to.

QUERY

```
SELECT
  a.Doctor_name, COUNT(DISTINCT p.patient_id) AS
  Count_of_UniquePatients
FROM
  patients AS p
  INNER JOIN appointments AS a
    ON p.patient_id = a.patient_id
GROUP BY
  a.doctor_name;
```

RESULT



	Doctor_name	Count_of_UniquePatients
▶	Alexis Butler	1
	Andrew King	1
	Andrew Sosa	1
	Benjamin Garcia	1
	Brian Shepherd	1

07

Ranking

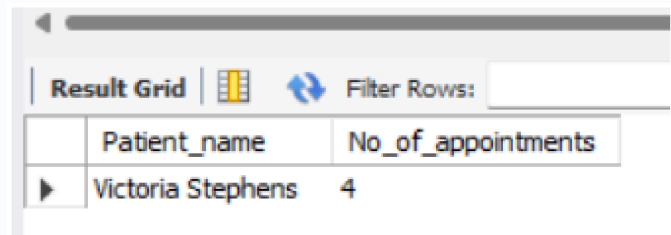
✓ Verified

Retrieve the name of the patient who has had the highest number of appointments.

QUERY

```
SELECT
  p.Patient_name, COUNT(a.appointment_id) AS
  No_of_appointments
FROM
  patients AS p
  INNER JOIN appointments AS a
    ON p.patient_id = a.patient_id
GROUP BY
  p.patient_name
ORDER BY
  COUNT(a.appointment_id) DESC
LIMIT 1;
```

RESULT



Patient_name	No_of_appointments
Victoria Stephens	4

08

Doctor Workload

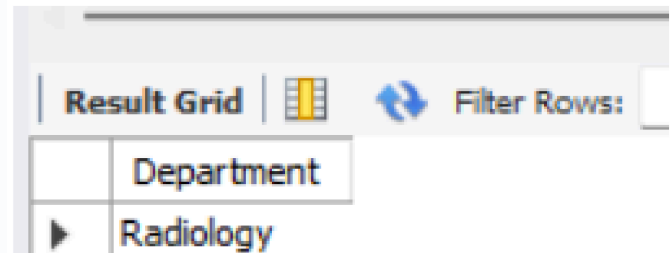
✓ Verified

Identify the department that has the most appointments scheduled.

QUERY

```
SELECT
  Department
FROM
  appointments
GROUP BY
  department
ORDER BY
  COUNT(appointment_id) DESC
LIMIT 1;
```

RESULT



Department
Radiology

09

Percentage Share

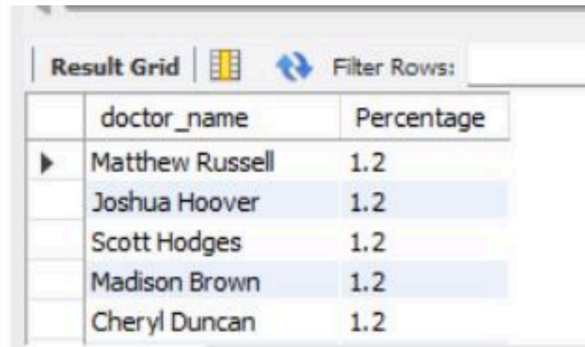
✓ Verified

Find the percentage of total appointments handled by each doctor.

QUERY

```
SELECT
    doctor_name,
    ROUND((COUNT(appointment_id) * 100) /
          (SELECT COUNT(*) FROM appointments), 1) AS
Percentage
FROM
    appointments
GROUP BY
    doctor_name;
```

RESULT



The screenshot shows a 'Result Grid' window with a table containing two columns: 'doctor_name' and 'Percentage'. There are five rows of data, each with a value of 1.2 in the 'Percentage' column. The doctors listed are Matthew Russell, Joshua Hoover, Scott Hodges, Madison Brown, and Cheryl Duncan. The window also includes a 'Filter Rows:' input field and a refresh icon.

doctor_name	Percentage
Matthew Russell	1.2
Joshua Hoover	1.2
Scott Hodges	1.2
Madison Brown	1.2
Cheryl Duncan	1.2